Internship Report

Forecasting the Exoplanet Yield and Observational Capabilities of the LIFE Mission

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Chapter 1

Introduction

1.1 Purpose

Serving as preparation for the Master's thesis to begin in March 2026 at University Tor Vergata Rome under the mentorship of Professor Amedeo Balbi

1.2 Background and Rationale

The detection and characterization of Earth-like exoplanets is one of the central goals of contemporary astrophysics. Direct detection missions are required to move beyond indirect discovery techniques and enable atmospheric characterization. The Large Interferometer for Exoplanets (LIFE) mission is a proposed space-based mid-infrared nulling interferometer designed to address this challenge. As the mission is still in its concept and simulation stage, understanding its potential scientific return depends on generating synthetic universes from prior catalogs of stars and planets, simulating the instrument's detection performance, and estimating detection yields under various assumptions. This internship provides the student with the opportunity to explore the theoretical and practical foundations of space interferometry, learn simulation tools used in mission design studies, and contribute to preparatory research relevant for future analyses in their Master thesis.

Chapter 2

Goals and Outcomes

The specific goals of the internship were:

- To acquire theoretical understanding of the LIFE concept, architecture, and science objectives.
- To review existing literature on exoplanet yield forecasting methodologies.
- To use the `LIFEsim` framework to simulate exoplanet observations.
- To visualize and interpret simulated planetary populations using Self-Organizing Maps (SOMs).
- To prepare a comprehensive report forming the basis for the upcoming Master's thesis.

By the end of the internship, the student is intended to gain familiarity with the LIFE simulation pipeline, executed yield-related simulations using `LIFEsim`, and developed visualization tools to analyze the resulting planetary populations. These outcomes establish a solid foundation for more advanced yield studies during the Master's thesis.

Chapter 3

Literature Summary

3.1 Literature

The literature relevant to the objectives of this internship primarily focuses on the detection and characterization of exoplanets using mid-infrared space interferometry.

Relevant literature reviewed as part of this internship are the following :

- Kammerer2018 J. Kammerer & S. P. Quanz, **"Simulating the exoplanet yield of a space-based mid-infrared interferometer based on Kepler statistics"**, A&A 609, A4 (2018). `https://arxiv.org/abs/1707.06820`
- Quanz2021 S. P. Quanz et al., **"LIFE I: Improved exoplanet detection yield estimates for a large mid-infrared space-interferometer mission"**, arXiv:2101.07500 (2021). `https://arxiv.org/abs/2101.07500`
- Glauser2024 A. M. Glauser et al., **"The Large Interferometer for Exoplanets (LIFE): a space mission for mid-infrared nulling interferometry"**, SPIE Proc. 13095 (2024). `https://orbit.dtu.dk/files/380743586/130951D.pdf`
- Angerhausen2023 D. Angerhausen et al., "**LIFE VIII: Where is the phosphine?"**, arXiv:2211.04975 (2022/2023). `https://arxiv.org/abs/2211.04975`
- Alei2024 E. Alei et al., **"LIFE XIII: The value of combining thermal emission and reflected light for the characterization of Earth twins**", A&A 664, A1 (2024). `https://arxiv.org/abs/2406.13037`
- Angerhausen2025 D. Angerhausen et al., "**What if we find nothing? Bayesian analysis of the statistical information of null results in future exoplanet habitability and biosignature surveys"**, AJ 169, 5 (2025). `https://arxiv.org/abs/2504.06779`

- Kammerer2022 J. Kammerer et al., **"LIFE VI: Detecting rocky exoplanets in the habitable zones of Sun-like stars"**, A&A 668, A52 (2022). `https://arxiv.org/abs/2210.01782`

3.2 Comparative Overview of LIFE Mission Studies

The seven studies summarized in this document investigate the scientific potential of the LIFE mission using different assumptions, parameter choices, and analysis methods. **Kammerer & Quanz (2018)** present the **first exoplanet yield estimates based on Kepler occurrence rates**, using simplified blackbody planet models and idealized instrument assumptions; the most important result is the **strong sensitivity of predicted yields to habitable-zone definitions**, exozodiacal dust levels, and interferometer aperture size. **Kammerer et al. (2022)** focus specifically on rocky planets in the habitable zones of Sun-like stars and introduce more realistic noise terms, detection thresholds, and target selection criteria, demonstrating that detecting Earth-sized planets remains feasible for a significant number of nearby targets. **Quanz et al. (2021)** improve yield predictions through end-to-end mission simulations that include updated occurrence rates, optimized interferometric baselines, and detailed mission architectures, showing that LIFE could detect dozens of terrestrial planets under realistic observing conditions.

Glauser et al. (2024) shift the emphasis from yield estimation to mission architecture and instrumental performance, identifying null depth, baseline stability, and formation-flying precision as key limiting parameters for LIFE. **Angerhausen et al. (2023)** investigate atmospheric characterization by modeling the detectability of phosphine (PH_3) for different planet types, atmospheric compositions, and integration times, highlighting the strong dependence of biosignature sensitivity on both planetary atmospheres and stellar environments. **Alei et al. (2024)** demonstrate the scientific value of combining LIFE thermal-emission spectra with reflected-light observations, showing that joint retrievals significantly improve constraints on planetary radius, temperature, and atmospheric composition compared to single-technique observations. Finally, **Angerhausen et al. (2025)** use Bayesian inference to assess the information content of null-result surveys, demonstrating that even non-detections can place strong statistical constraints on planet occurrence rates, with results depending critically on the assumed prior distributions and number of observed targets.

Detailed literature summary outlining current knowledge and key assumptions can be found on the following link:

`https://github.com/SonaDavis-03/Internship-LifeSim-/blob/main/INTERNHSIP_Literature_SUMMARY.pdf`

Chapter 4

Simulation Setup

4.1 LIFEsim Framework

The primary simulation tool used during the internship was `LIFEsim`, a Python-based framework developed for simulating the performance of the LIFE mission. `LIFEsim` models the instrument response, noise sources, and observational strategy to generate synthetic mid-infrared spectra for exoplanets drawn from an input catalog.

4.2 Input Catalog and Planet Classification

The simulations performed during this internship are based on a predefined example catalog containing synthetic planetary systems designed for yield and performance studies. Each catalog entry corresponds to a single planet–star system realization and includes orbital, physical, and observational parameters required by the `LIFEsim` framework.

The catalog is structured to support Monte Carlo sampling through the inclusion of a universe index (`nMC` or `Nuniverse`), allowing multiple realizations of planetary systems under identical statistical assumptions. Planetary parameters include radius, mass, orbital period, eccentricity, inclination, and orbital phase angles. Stellar properties such as stellar radius, mass, effective temperature, distance, and spectral type are also provided.

In addition, the catalog contains derived observational quantities, including angular separation, maximum angular separation, incident flux, planetary equilibrium temperature, and assumed levels of exozodiacal dust. These parameters enable a direct assessment of observability within the LIFE mission simulation pipeline.

Table 4.1 lists a representative subset of parameters included in the catalog together with example values for a single planet–star system.

Table 4.1: Representative parameters included in the example exoplanet catalog used for `LIFEsim` simulations. The table shows example values for one catalog entry.

Parameter	**Description**	**Example Value**
nMC / Nuniverse	Monte Carlo universe index	0
R_p	Planet radius ($R_\oplus$)	0.832
P_{orb}	Orbital period (yr)	0.773
M_p	Planet mass ($M_\oplus$)	0.347
e_p	Orbital eccentricity	0.000
i_p	Orbital inclination (rad)	1.497
A_{bond}	Bond albedo	0.033
z	Exozodiacal dust level	0.713
a_p	Semi-major axis (AU)	0.0165
θ_p	Orbital phase angle (rad)	3.779
T_p	Planet equilibrium temperature (K)	2141
R_s	Stellar radius ($R_\odot$)	1.01
M_s	Stellar mass ($M_\odot$)	1.00
T_s	Stellar effective temperature (K)	5720
D_s	Stellar distance (pc)	14.93
Spectral type	Host star spectral class	G
Target name	Host star identifier	*20 LMi

Planets were classified as habitable or non-habitable based on their orbital distance and equilibrium temperature, following commonly adopted criteria in the literature. This classification was used in subsequent analyses to compare detectable populations and investigate the ability of LIFE observations to distinguish between different classes of terrestrial exoplanets.

4.3 Spectral Simulations

For each planet in the catalog, `LIFEsim` was used to simulate mid-infrared spectra under assumed instrument configurations. The simulations accounted for instrumental noise and astrophysical background contributions. The resulting spectra formed the basis for yield estimation and further analysis.

4.4 Yield Estimates

The `LIFEsim` framework can be used to estimate exoplanet detection yields for the search phase of the LIFE mission. In this context, yield is defined as the number of planets

that satisfy the instrument detection criteria under assumed mission parameters and astrophysical priors.

To ensure reasonable computation times during exploratory analysis, a reduced exoplanet population catalog was used. This catalog was generated using the P-Pop framework and contains a representative subset of planetary systems suitable for yield demonstrations. While simplified, this approach captures the key dependencies between stellar type, planetary properties, and detectability. An example of a full catalog is available in the LIFE documentation.

As a first demonstration, the reduced P-Pop-generated exoplanet catalog was ingested into `LIFEsim` to investigate which regions of parameter space are accessible to LIFE observations. Each simulation consists of multiple Monte Carlo universes, allowing statistical variations in detection outcomes to be explored.

The parameters used in this analysis include:

- Stellar spectral type (M, K, G, F)
- Stellar distance, restricted to targets within 10 pc
- Planetary radius and mass
- Orbital period and semi-major axis
- Planetary equilibrium temperature
- Detection flag returned by `LIFEsim`

4.5 Self-Organizing Maps

To visualize and interpret the multidimensional simulation output, Self-Organizing Maps (SOMs) were employed. SOMs provide a low-dimensional representation of high-dimensional data, enabling the identification of clusters and trends within the simulated exoplanet population. Spectral features and planetary parameters were used as inputs to the SOM.

Chapter 5

Results

The simulation results focus on the characterization and visualization of exoplanets deemed observable by the LIFE mission under the assumptions implemented in the `LIFEsim` framework. Rather than providing a single scalar yield value, the analysis emphasizes the properties of the detectable planetary population and the information content of their simulated spectra.

5.1 Example of a Single Expected Observation

To illustrate the outcome of the simulation pipeline, a representative planet from the detectable sample was selected. Figure 5.1 shows the simulated planetary flux as a function of wavelength in the mid-infrared, together with the corresponding signal-to-noise ratio (SNR) per wavelength bin.

The flux spectrum reflects the thermal emission of the planet and contains molecular absorption features relevant for atmospheric characterization. The SNR curve demonstrates how detectability varies across the spectral range, highlighting wavelength regions where LIFE is most sensitive.

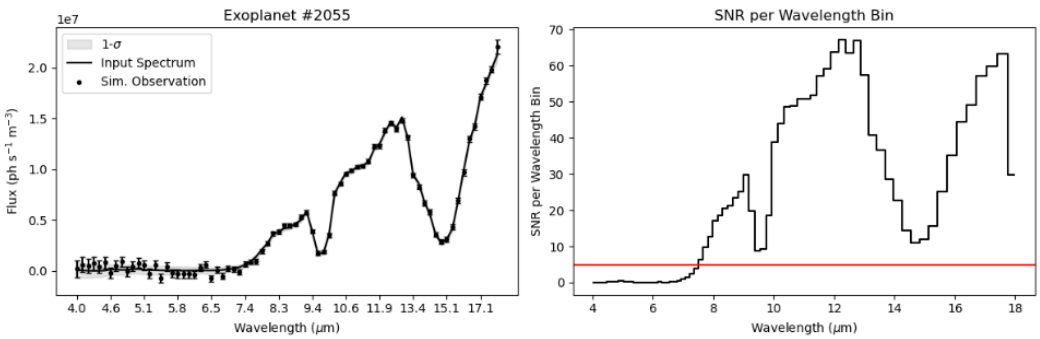

Figure 5.1: Simulated observation of a representative exoplanet using LIFEsim. Top panel: planetary flux as a function of wavelength. Bottom panel: signal-to-noise ratio per wavelength bin.

5.2 Yield Estimates

5.2.1 Detectable Parameter Space of Exoplanets

LIFEsim can be used to estimate exoplanet yields for the search phase of LIFE. As a first demonstration, the reduced P-Pop-generated exoplanet catalog was ingested into `LIFEsim` to investigate which regions of parameter space are accessible to LIFE observations. Each simulation consists of multiple Monte Carlo universes, allowing statistical variations in detection outcomes to be explored.

Figure 5.2 shows the mean number of detected planets per stellar type, averaged over multiple Monte Carlo universes. Error bars indicate the statistical spread between universes.

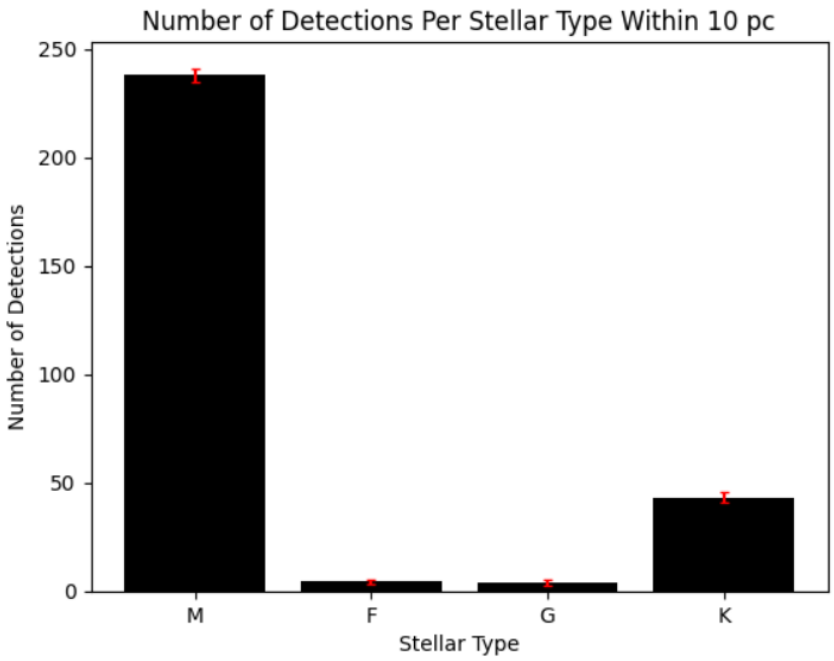

Figure 5.2: Mean number of detected exoplanets per stellar type within 10 pc, averaged over multiple Monte Carlo universes. Error bars represent the standard deviation between universes.

The results indicate that M-type stars dominate the detection yield, primarily due to their favorable planet-to-star contrast and smaller habitable zone separations, which are well matched to the LIFE interferometric baseline.

5.2.2 Habitable-Zone Planets Around M-type Stars

To further illustrate LIFE's sensitivity to potentially habitable planets, a single Monte Carlo universe was examined in detail, focusing on habitable-zone planets orbiting M-type stars. This analysis highlights the subset of planets that are both detectable and located within the classical habitable zone.

Figure 5.3 shows the distribution of habitable-zone planets around M stars for a representative universe, demonstrating the expected contribution of low-mass stars to LIFE's terrestrial planet yield.

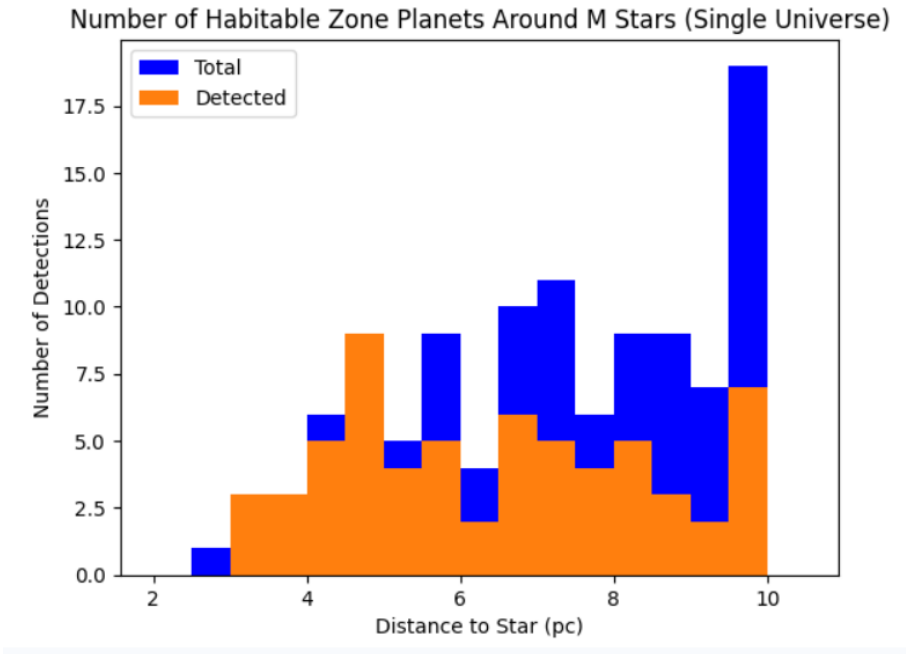

Figure 5.3: Detected habitable-zone exoplanets orbiting M-type stars for a representative Monte Carlo universe.

The obtained dominance of M-type stars is consistent with Kammerer & Quanz (2018) and Quanz et al. (2021), who also found that detection yield is strongly enhanced for late-type stars due to improved contrast and smaller habitable zone separations. These yield estimates provide an initial, catalog-dependent assessment of LIFE's detection capabilities and motivate the more detailed spectral and population analyses presented in the subsequent sections.

5.3 Average Spectra per SOM Cell

After projecting the simulated planets onto the Self-Organizing Map, average spectra were computed for each SOM cell. Figure 5.4 visualizes these mean spectra, enabling a comparison between different regions of the map.

The first panel shows the NaN-aware Self-Organizing Map trained using a combination of spectral data and physical parameters. This representation highlights the topology of the SOM and the distribution of planets across the grid. The second panel displays the average simulated spectra for each SOM cell, enabling a direct comparison between different regions of the map.

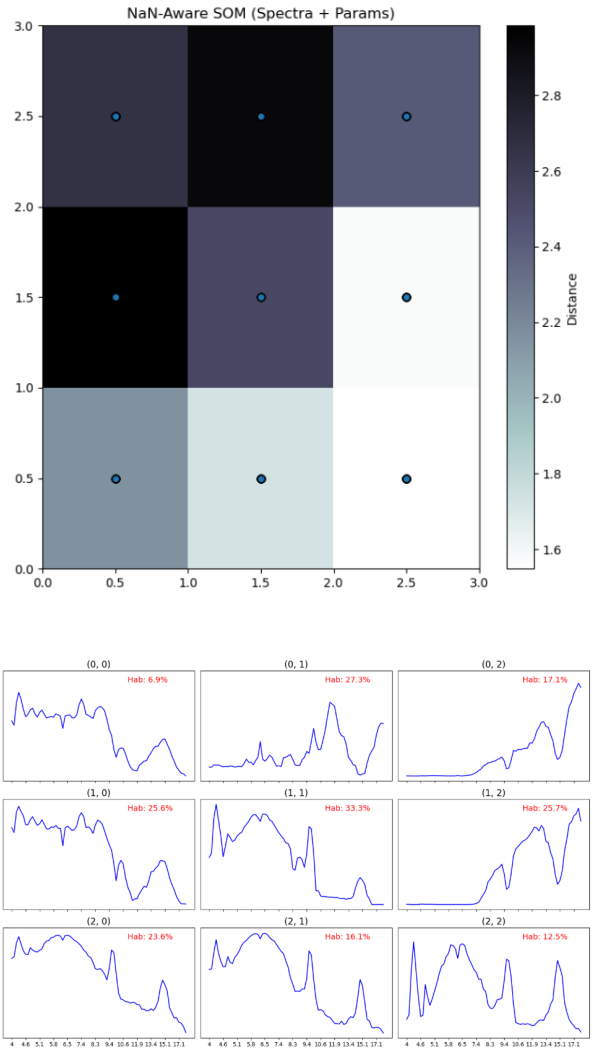

Figure 5.4: Self-Organizing Map analysis of simulated exoplanet spectra. Top: NaN-aware SOM trained using spectral and physical parameters. Bottom: Average spectra associated with each SOM cell.

5.4 Parameter Distributions Across the SOM Grid

To better understand the physical meaning of the SOM organization, key planetary parameters such as radius, equilibrium temperature, and orbital distance were visualized across the grid. Figure 5.5 shows how these parameters vary spatially over the SOM.

The smooth gradients observed for several parameters indicate that the SOM preserves meaningful relationships between physical properties and spectral characteristics.

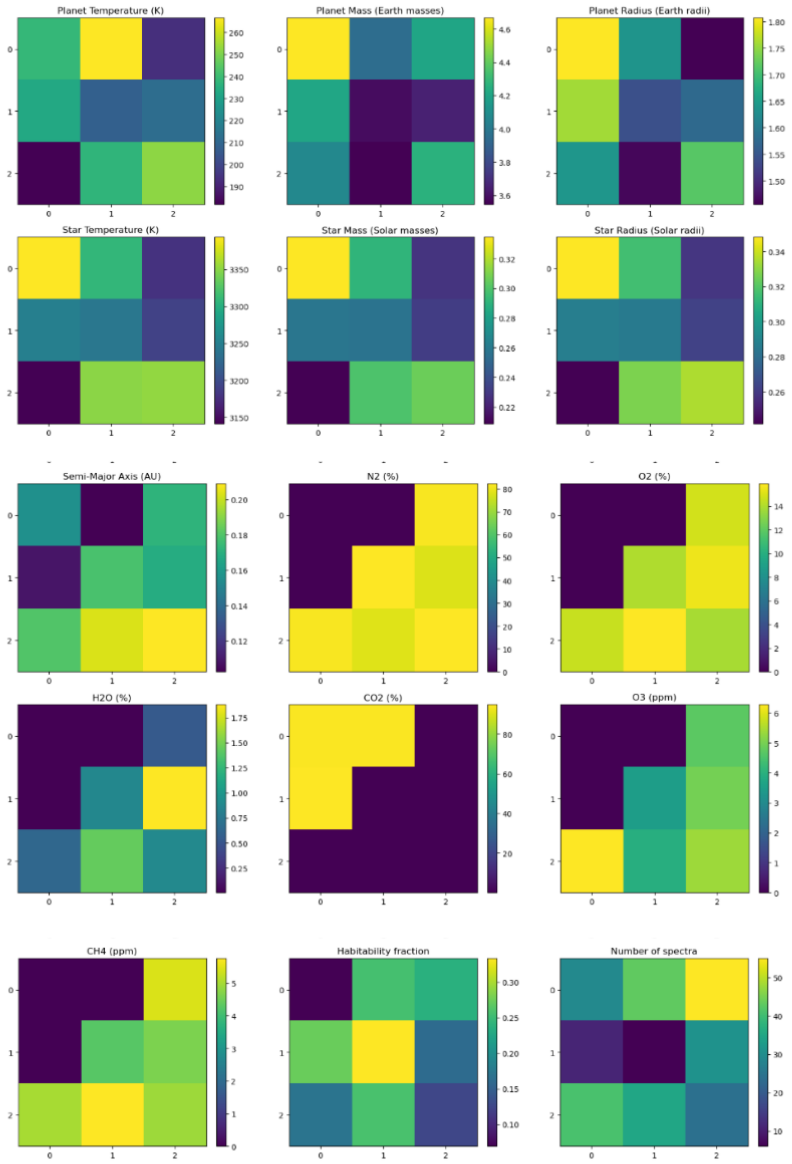

Figure 5.5: Distribution of selected planetary parameters across the Self-Organizing Map grid. Color scales indicate average parameter values within each cell.

5.5 Habitable versus Non-Habitable Planet Populations

Finally, the distributions of key parameters were compared between planets classified as habitable and non-habitable in the input catalog. Figure 5.6 summarizes these differences.

The analysis shows clear separations in temperature, orbital distance, and spectral

characteristics, suggesting that LIFE observations combined with SOM-based analysis can effectively distinguish between these two populations.

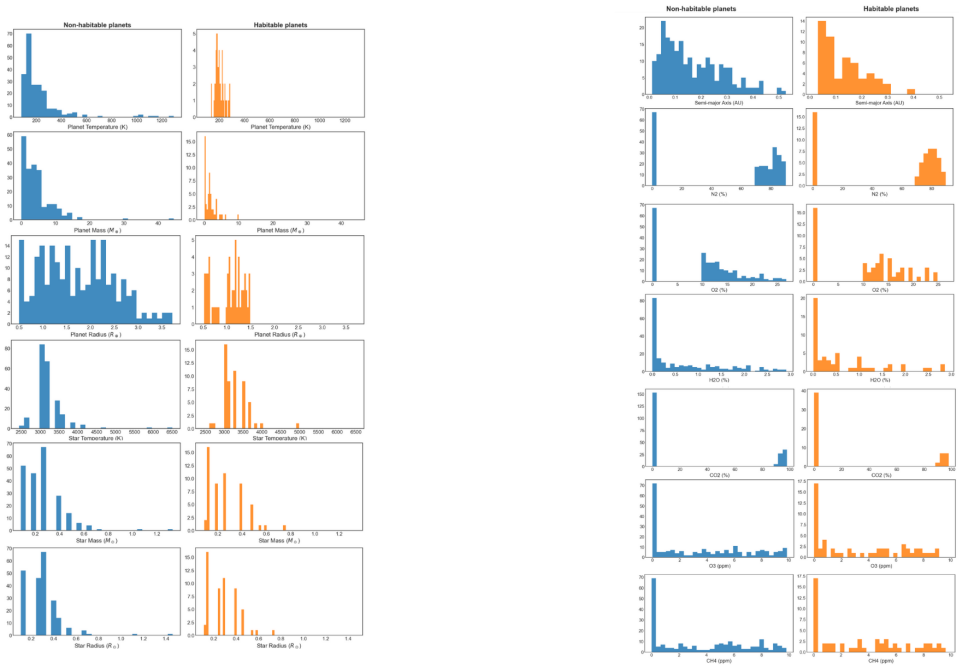

Figure 5.6: Comparison of parameter distributions for habitable and non-habitable exoplanets in the simulated sample.

The simulations produced a set of synthetic spectra corresponding to planets in the input catalog that are potentially observable by LIFE. The yield, in this context, is defined as the subset of catalog planets for which `LIFEsim` predicts feasible spectral observations under the assumed mission parameters.

Visualization using SOMs revealed clear clustering between habitable and non-habitable planets in parameter space. Histograms of planetary and stellar properties showed systematic differences between these populations, particularly in temperature, orbital distance, and atmospheric composition. These results demonstrate the ability of LIFE to not only detect planets but also to distinguish between different classes of terrestrial exoplanets.

All simulation notebooks developed and used during this internship are publicly available in a dedicated GitHub repository. The repository contains the full `LIFEsim`-based workflow, data preprocessing steps, SOM training and visualization scripts, and figure-generation notebooks corresponding to the results presented in this report.

`https://github.com/SonaDavis-03/Internship-LifeSim-`

Chapter 6

Discussion and Conclusion

The Large Interferometer for Exoplanets (LIFE) represents a groundbreaking initiative in the quest to detect and characterize terrestrial exoplanets in the habitable zone The internship results highlight the strong dependence of predicted yield on both instrument performance and astrophysical assumptions. While the present simulations were limited to a fixed input catalog and baseline configuration, they illustrate how tools such as `LIFEsim` can be used to explore trade-offs and sensitivities.

The use of SOMs proved particularly valuable for interpreting complex, high-dimensional outputs and will be further developed during the Master's thesis.

This internship provided essential analysis of the capabilities of LIFE mission. Through hands-on use of `LIFEsim` and advanced visualization techniques, practical experience in exoplanet yield forecasting and mission performance analysis was gained.This work demonstrates that LIFE has strong potential to detect and characterize terrestrial planets within 10 pc, particularly around M-type stars. The combination of Monte Carlo yield forecasting and SOM-based spectral analysis provides a powerful framework for optimizing future mission design.

The work completed during the internship establishes a strong conceptual and technical foundation for future research aimed at refining yield estimates and optimizing the scientific return of the LIFE mission.

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